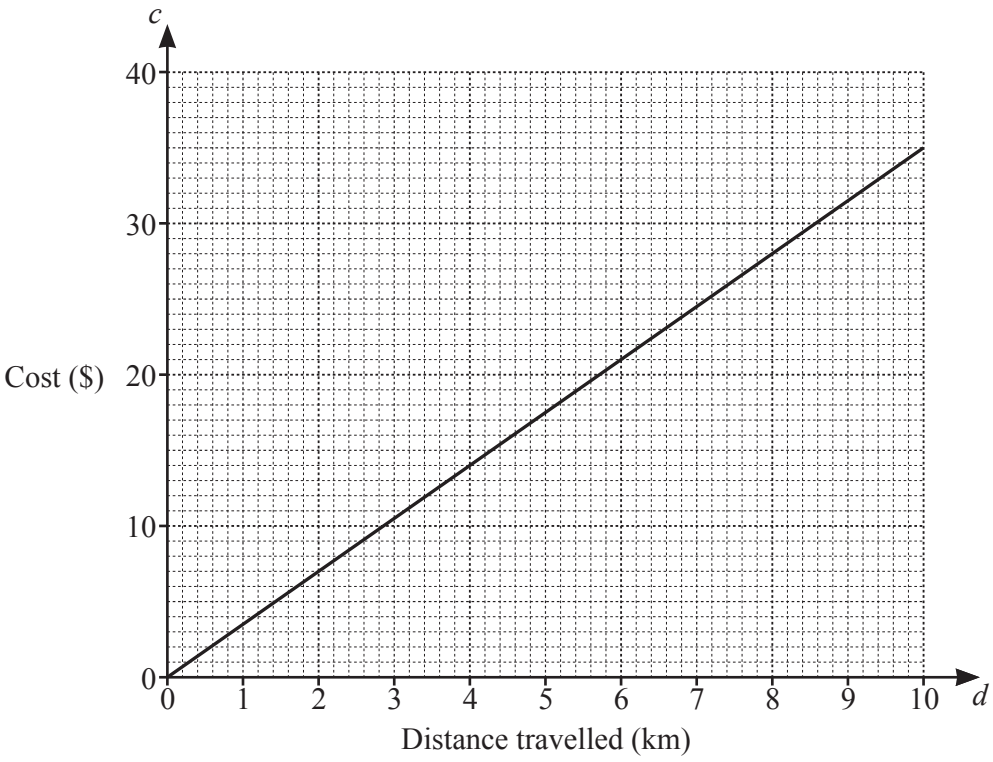
0580_m20_qp_32_Q5

5 (a)



(i) The graph shows the cost, $\$c$, of travelling a distance, d km, with *Saanvi's Taxis*.

(a) Write down the cost of a 4 km journey.

$ ..... [1]

(b) Complete this statement.

Saanvi's Taxis cost $ ..... for each kilometre travelled. [1]

(c) Find the equation of the line.

$c =$ ..... [1]

(ii) *Krishna's Taxis* cost $5 to hire plus $2 for each kilometre travelled.

(a) Show that the cost of a 4 km journey with *Krishna's Taxis* is $13.

[1]

(b) Find an equation for the cost, $\$c$, of travelling d kilometres with *Krishna's Taxis*.

$c =$ ..... [2]

(c) On the grid, draw a line to show the cost of travelling with *Krishna's Taxis*. [2]

(d) Mrs Singh wants to hire a taxi.
She says that *Saanvi's Taxis* are always cheaper than *Krishna's Taxis*.

Is Mrs Singh correct?
Give a reason for your answer. Use your graph to help you.

..... because .....

..... [1]

(b) A minibus can be hired from *Dhruv's Minibuses*.
The cost is \$$h$ per hour plus \$$p$ per passenger.

(i) When the minibus is hired for 3 hours with 10 passengers the cost is \$61.

Complete the equation.

$$3h + 10p = \text{.....}$$

[1]

(ii) When the minibus is hired for 5 hours with 8 passengers the cost is \$80.

Write this information as an equation.

..... = ...

[2]

(iii) Solve your two simultaneous equations to find h and p.
You must show all your working.

$h =$ .....

$p =$ ..... [4]